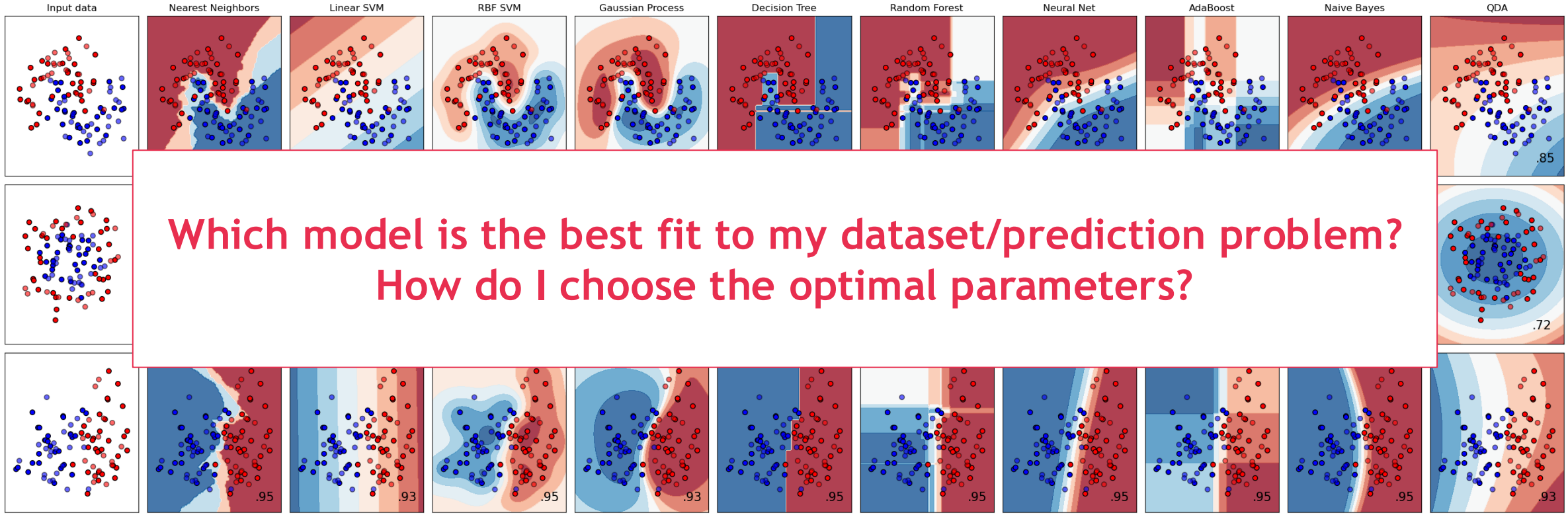
05

Model Selection

Classifiers



Which model is the best fit to my dataset/prediction problem?
How do I choose the optimal parameters?

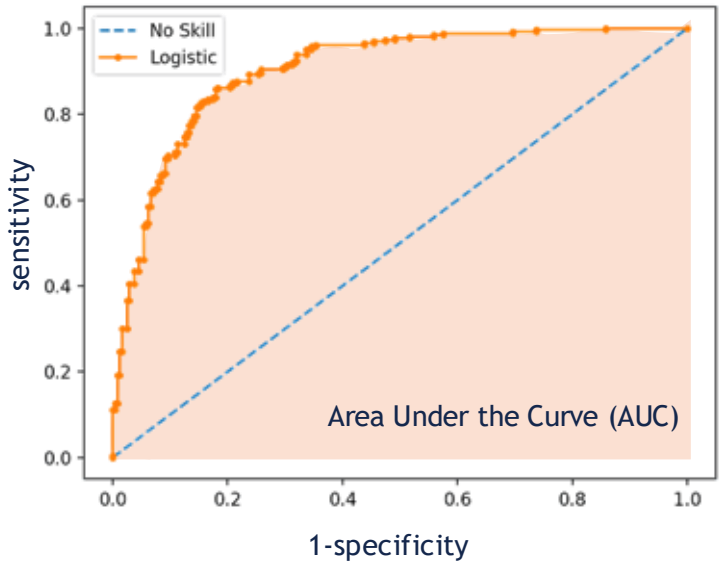
Model selection

- The goal of machine learning is **generalization, not memorization**
- Different models might perform differently depending on
  - Data size
  - Data noise
  - Feature dimensionality
  - (non-)Linearity
- Model selection = choosing the best model + hyperparameters that yield the **best expected performance on unseen data**

Evaluation metrics (classification)

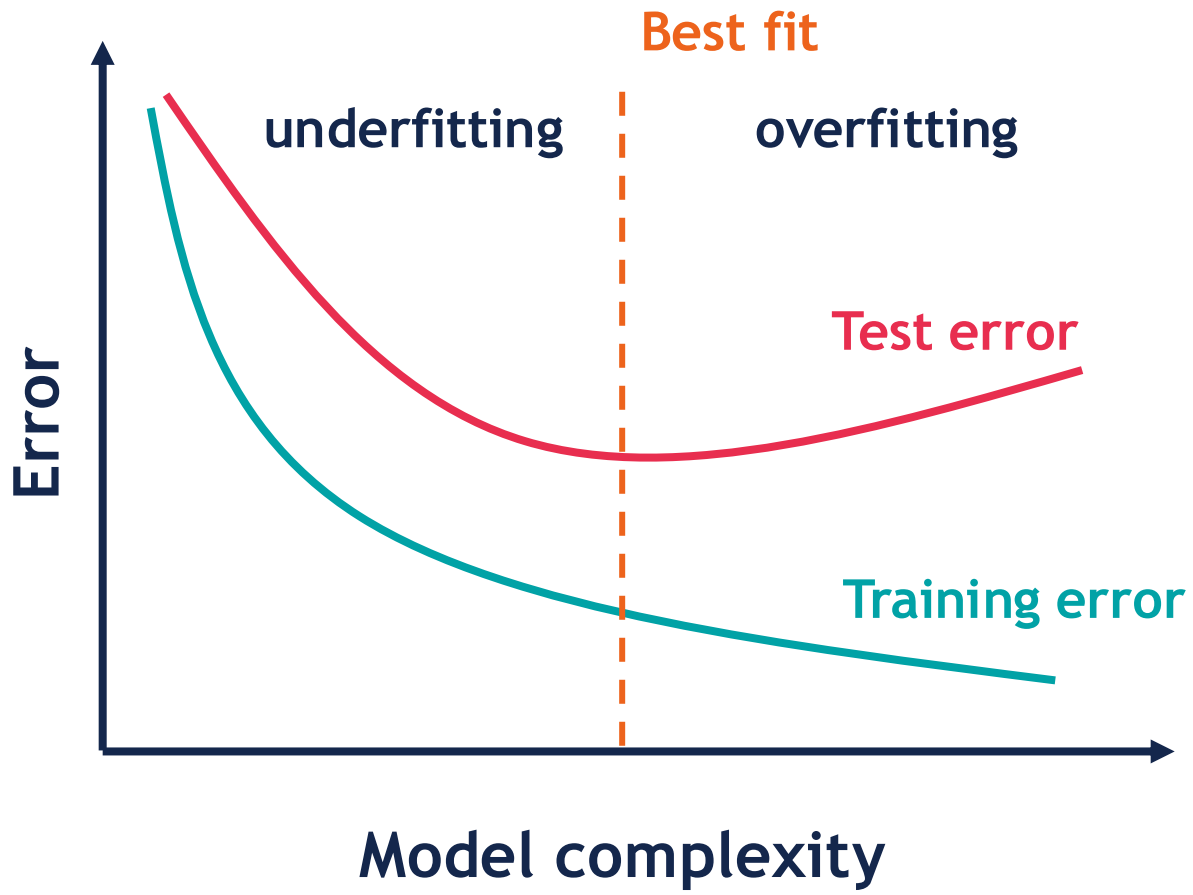
- Accuracy – only useful for balanced datasets
- Precision, Recall, F1-score – useful for imbalanced data
- ROC-AUC – threshold-independent metric
- Confusion matrix – interpretable breakdown *(not a metric on its own)*

The "optimal" model will depend on your choice of the evaluation metric



Really is … \ Classified as …	*positive*	*positive*
positive	TP	FN
negative	FP	TN

Train-test splitting



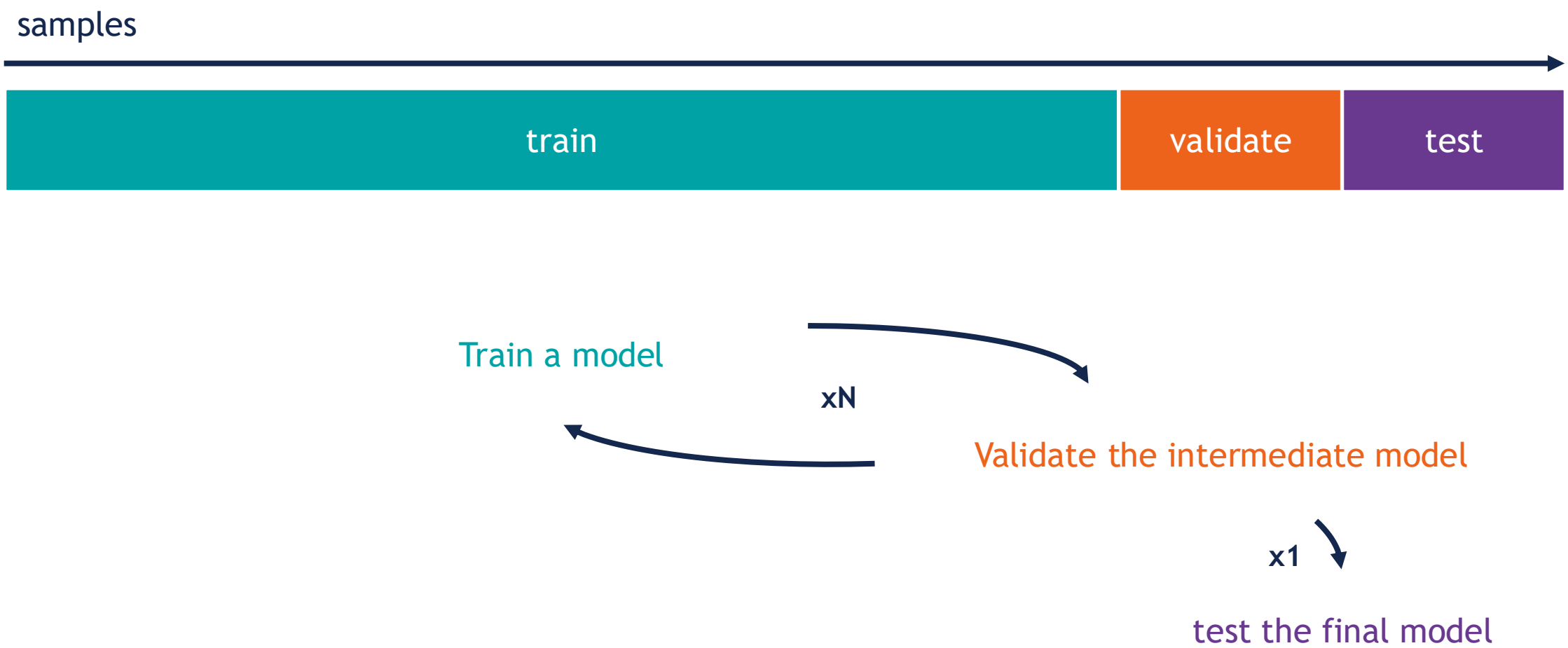
- Test set = **independent dataset** to measure model performance
- Model learns on training data
- Evaluated on (unseen) test data

→ Good generalization to new inputs

VIB

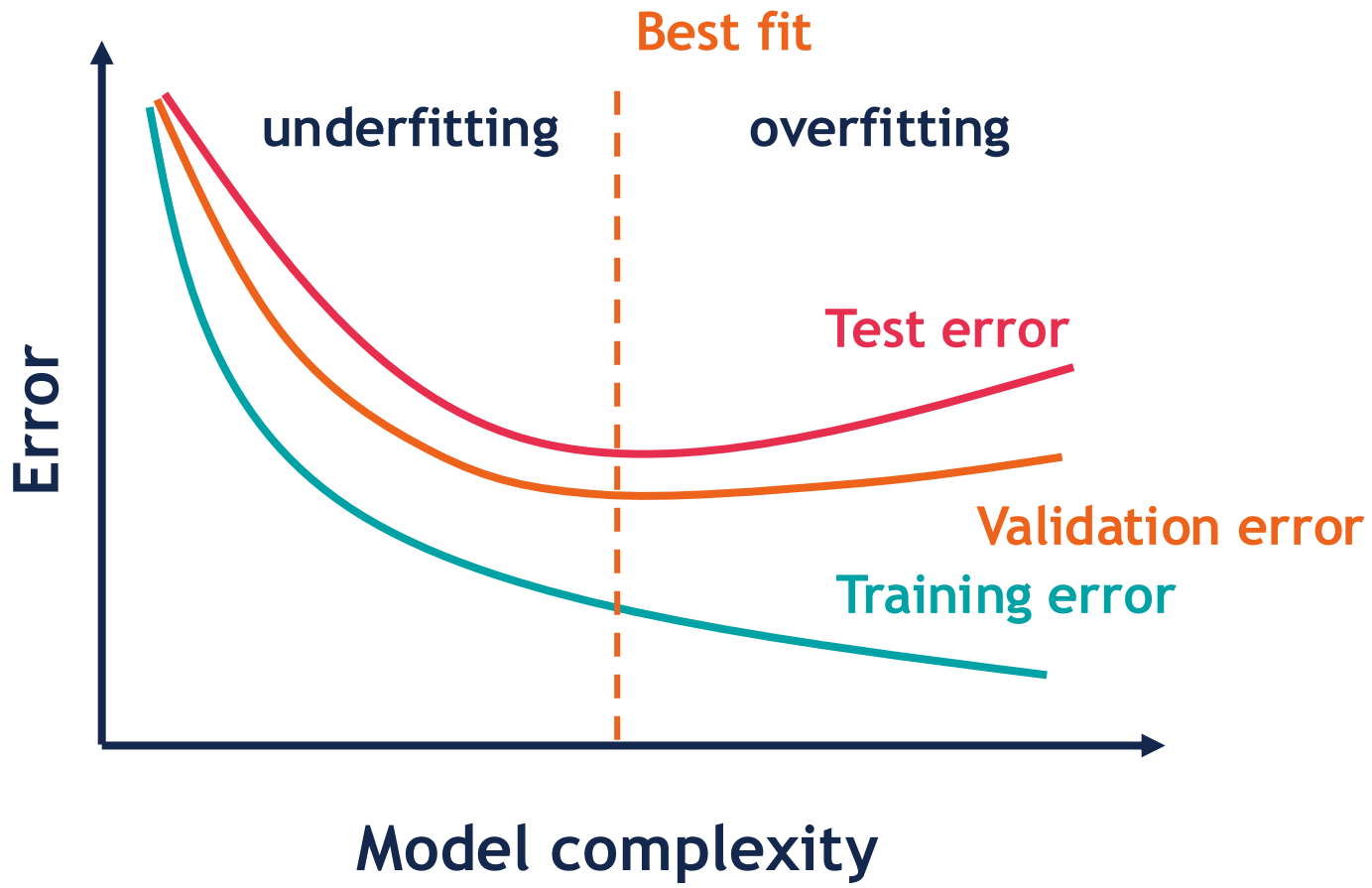
How do we compare the performance of different models
without consuming the test data?

Train-test splitting



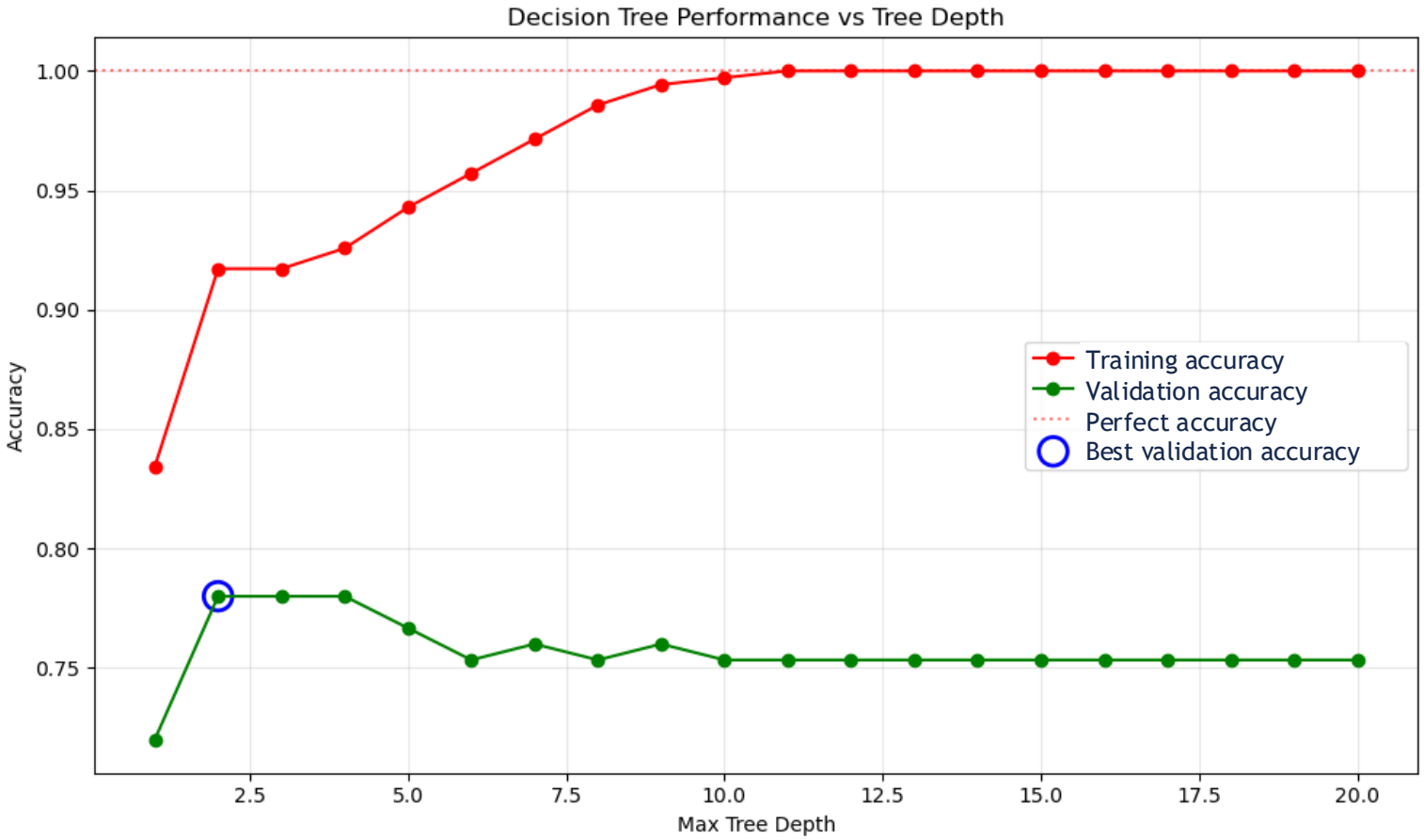
VIB

Train-test splitting



- Test set can only be used once!
- The validation data models the aspect of “unseen data”
- Model selection involves optimization on the validation error

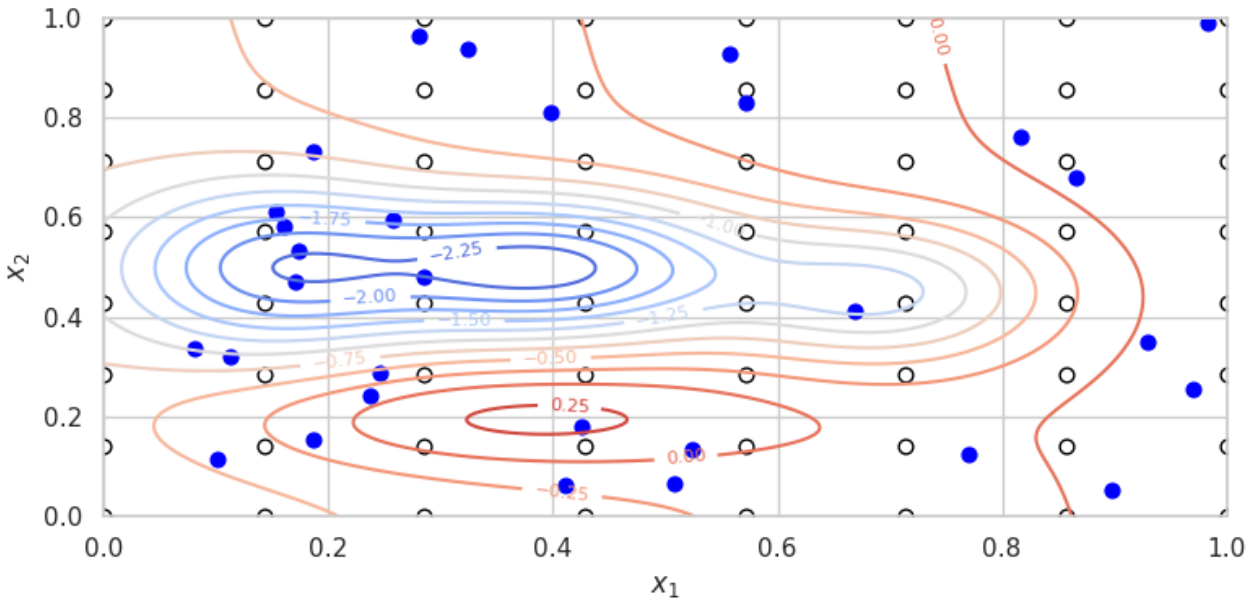
Hyperparameter tuning



Hyperparameter tuning

Optimization on the validation error *(similar to loss optimization during training, but now across hyperparameters)*

- Grid Search
  - Systematic but expensive
  - Best for low-dimensional hyperparameter spaces
- Random search
  - Surprisingly effective
  - Much more efficient for large spaces



Cross-validation

With extensive optimization, the model might overfit on the validation set

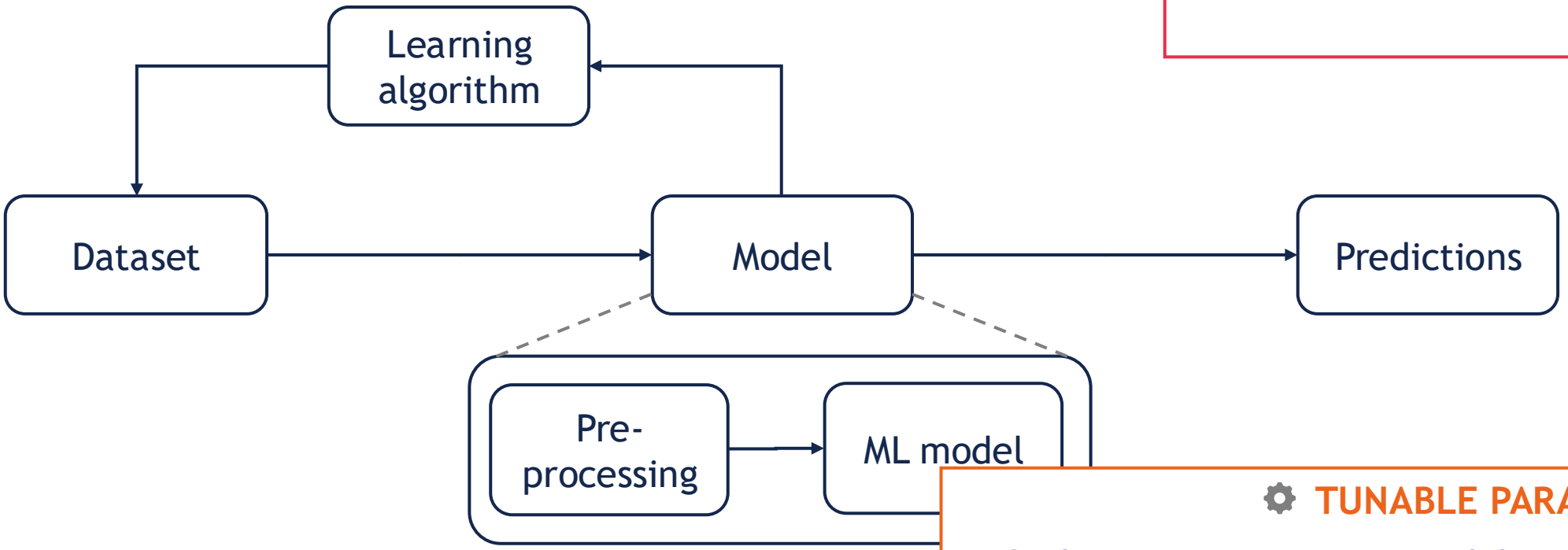
Cross-validation

- Iterative optimization on separate train/validation splits
- Reduces variance in the performance estimation

Machine Learning Pipeline

MODEL SELECTION

- Cross-validation
- GridSearch/ RandomSearch



TUNABLE PARAMETERS

- Scaling/imputation strategy
- Dimensionality reduction
- Model
- Regularization
- **Hyperparameters**

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